

Alejandro Soler González

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Physicist (M.Sc. Space Systems Engineering, **GPA 9.1/10**) targeting **Quantitative Research / Trading**: built and ran a live cross-venue execution engine in **Rust**, trading real capital on prediction markets from colocated cloud infrastructure. Available full-time from September 2026.

EDUCATION

M.Sc. in Space Systems Engineering — Universidad Europea, Madrid 2025 – Sep 2026 (expected)
GPA 9.1/10

- **Master's Thesis** (in progress): real-time trajectory optimization for autonomous rocket landing — posed the final 2 km of powered descent as a sequence of **convex programs (SOCP)** solved in closed loop inside a self-built **3-/6-DoF simulator** — the same convex-guidance method family used for SpaceX booster landings. [Interactive demo](#).

B.Sc. in Physics — Universidad Complutense de Madrid (bilingual Spanish/English program) 2021 – 2025

- **Bachelor's Thesis (9.2/10)**: multi-agent **Bayesian search** — coordinated agents locate the maximum of an unknown field via **Gaussian-process (Kriging) regression**, balancing exploration vs. exploitation under uncertainty. Selected for Docta Complutense (UCM's distinguished-thesis repository). [Interactive demo](#).
- Coursework: mathematical methods, statistical physics, scientific computing, probability & statistics; quantum-computing program (3 algorithms implemented in **Qiskit**).

EXPERIENCE & PROJECTS

Independent Trader — Cross-Venue Bitcoin Structure (Polymarket / Kalshi) Feb – May 2026

- Found and traded a predictable settlement gap between **Kalshi and Polymarket** on hourly Bitcoin: their two price oracles disagree by the **USDT/USD stablecoin basis** — signed and forecastable before each hour resolves. Wrote up the mechanism as a public research note, validated on 1,350 real Kalshi settlements.
- Built the execution engine in **Rust** (chosen over Python for latency), running 24/7 on **AWS eu-west-1 (Ireland)** for proximity to exchange infrastructure; automated order placement, fill monitoring and logging. Prototyped with two partners, then **rebuilt the entire system solo**.
- Managed **single-leg ("hanging-leg") execution risk** and cross-platform resolution risk with hard position caps; wound down operations when Spain restricted both venues (May 2026).

Flight Dynamics Intern — PLD Space (launch-vehicle company, Elche, Spain) Mar – Apr & Jun – Aug 2025

- Built **MATLAB** tools automating recurring flight-dynamics analyses for the team; reworked the **finite-difference surface discretization** used to compute panel areas, improving the accuracy of pressure-to-force integration on the launch vehicle.

Co-founder & Sole Developer — Bolbee, AI marketing platform for SMEs 2025 – present

- Sole engineer of the full product (Next.js, Supabase/SQL, Stripe, multi-model AI pipeline); 10 active beta users, market launch summer 2026; own architecture, infrastructure and integrations end-to-end.

Team Lead — Exoplanet Detection with ML, NASA Space Apps Challenge (Madrid) 2025

- Led a 3-person team building a neural-network exoplanet classifier on NASA telescope data (**Python**): owned data cleaning and feature selection, defined the train/test methodology and evaluation (confusion matrix); delivered a working model in 48 h.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas; numerical methods, data/eval pipeline of the NASA project), Rust (trading execution engine), MATLAB (flight dynamics, simulators), SQL, TypeScript.

Quantitative: convex optimization, probability & statistics, Gaussian processes, Monte Carlo methods, control & estimation theory, numerical optimization.

Tools: Linux (server administration), Git, AWS, REST APIs.

LANGUAGES

Spanish (native) | English (C1, Cambridge certified) | Catalan (C1)

HONORS & ADDITIONAL

- Merit scholarship: full first-year university tuition (ranked 1st in graduating high-school class).
- Skipper's license (PER, vessels up to 24 m).